

# Lectures 5 and 6

# Lecture 5

- recap: intuitive method
- unfinished Example 4 from Lecture 4 Slides
- Lagrangians
- optimal bundles with concave preferences (Example 5 from Lec 4 slides)
- possibly start (non-mathy) demand properties

## Recap: Intuitive Method

- assume convex preferences: if  $x_1 \uparrow$  and  $x_2 \downarrow$ , then  $\frac{MU_1}{MU_2} \downarrow$
- how do you find the utility-maximizing bundle  $(x_1, x_2)$  among feasible choices, namely those that are:
  - non-negative:  $x_1 \geq 0$  and  $x_2 \geq 0$
  - affordable:  $p_1 x_1 + p_2 x_2 \leq m$

## Recap: Intuitive Method

- find the bundle that:
  - spends all your \$, i.e.  $p_1x_1 + p_2x_2 = m$
  - allocates \$ optimally, by equating bang-per-buck:  $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$
- caveat 1: if the above math gives you  $x_2 < 0$ , or if all bundles have  $\frac{MU_1}{p_1} > \frac{MU_2}{p_2}$ , go for the corner  $(x_1, x_2) = \left(\frac{m}{p_1}, 0\right)$
- caveat 2: if the above math gives you  $x_1 < 0$ , or if all bundles have  $\frac{MU_2}{p_2} > \frac{MU_1}{p_1}$ , go for the corner:  $(x_1, x_2) = \left(0, \frac{m}{p_2}\right)$
- caveat 3: if you hit issues like non-differentiability, use logic for the optimal allocation condition: eg perfect complements replace it by  $x_1 = x_2$

# Revisit Lecture 4

- example 4 (not done at all last class)
- briefly, examples 3 and 5 (done last class but rushed)

## Intuitive Method: Example 4

- now take  $u(x_1, x_2) = x_1 + 2x_2\sqrt{x_1} + x_2^2$ , with  $p_1 = 1, p_2 = 4$ , income  $m$
- here, if you use this utility function directly, you get

$$\frac{MU_1}{MU_2} = \frac{1 + \frac{2x_2}{2\sqrt{x_1}}}{2\sqrt{x_1} + 2x_2} = \frac{1}{2\sqrt{x_1}}$$

(or notice  $u(x_1, x_2) = (\sqrt{x_1} + x_2)^2$ , and work with its square root, which is a monotone transformation)

- these are convex: as  $x_1 \uparrow$  and  $x_2 \downarrow$  (good 1 relatively more abundant),  $\frac{MU_1}{MU_2}$  falls (good 1 relatively less valuable)
- so let's look for a point on the budget line where the two goods have equal bang-per-buck

## Example 4 Cont'd

- given  $p_1 = 1$  and  $p_2 = 4$ , on the budget line means

$$x_1 + 4x_2 = m \quad (1)$$

- and since we just calculated  $\frac{MU_1}{MU_2} = \frac{1}{2\sqrt{x_1}}$ , the equal bang-per-buck condition is:

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2} \Rightarrow \frac{1}{2\sqrt{x_1}} = \frac{1}{4} \Rightarrow \sqrt{x_1} = 2 \Rightarrow x_1 = 4 \quad (2)$$

- plug (2) into (1) to get  $x_2 = \frac{m-4}{4}$
- as long as  $m \geq 4$ , this is feasible so we have the solution:  
 $(x_1, x_2) = \left(4, \frac{m-4}{4}\right)$

## Example 4 Cont'd

- possible problem: what if  $m < 4$ ?
- here, your math told you  $x_1 = 4$  and  $x_2 = \frac{m-4}{4}$ , which is negative (not allowed)
- quick-and-dirty solution: buy zero of the good where your math gives you a negative quantity
- so optimal bundle is:

$$x_2 = 0, x_1 = \frac{m}{p_1} = m$$



## Intuitive Method: Example 4 Cont'd

- logic behind quick-and-dirty solution: if math gives you  $x_i < 0$ , it means that at ANY feasible point, good  $i$  has less bang-per-buck, so buy the other good only
- here:
  - feasibility implies  $x_1 \leq \frac{m}{p_1}$  (otherwise you spend more than income on good 1, impossible with  $x_2 \geq 0$ )
  - so if  $p_1 = 1$  and  $m < 4$ , then feasibility  $\Rightarrow x_1 < 4$ , and so

$$\frac{MU_1}{MU_2} = \frac{1}{2\sqrt{x_1}} > \frac{1}{4}$$

$$\text{while } \frac{p_1}{p_2} = \frac{1}{4}$$

- Thus at income levels  $m < 4$ , good 1 has more bang-per-buck at any feasible point
- In other words, we're in the caveat case where good 1 always has more bang-per-buck, so spend all \$ on it

## Quick Recap of Example 3: Perfect Complements

- with  $u(L, R) = \min\{L, R\}$ , intuitively it's optimal to set  $L = R$ 
  - why waste money on extra shoes of one type that give you no utility?
- plugging this into budget line  $p_L L + p_R R = m$ , the optimal bundle is  $L = R = \frac{m}{p_L + p_R}$
- you *could* get here also by comparing bang-per-bucks, if you understand what to do when equating is impossible:
- here,  $MU_L = 0$  if  $L > R$  (no value for extra left shoes) and  $MU_R = 0$  if  $R > L$
- so  $\frac{MU_L}{MU_R} < \frac{p_L}{p_R}$  if  $L > R$ , meaning left shoes have less bang-per-buck if  $L > R$ , which is a nudge to reduce  $L$
- and  $\frac{MU_R}{MU_L} < \frac{p_R}{p_L}$  if  $R > L$ , which is a nudge to reduce  $R$  (they have less bang-per-buck) when  $R > L$
- so having extra of one type is never optimal, leaving  $L = R$  as the only possibility

## Method #2: Lagrangians

- this is the formal math behind the intuitive method
- we'll see that really, the intuitive method simply jumps to the step of solving the Lagrangian FOC's

# General Recipe

- Suppose you want to choose  $x = (x_1, x_2, \dots, x_n)$  to maximize  $f(x)$ , subject to some inequality constraints, eg  $g_i(x) \geq 0$  or  $g_i(x) \leq 0$ 
  - for us, it's choosing  $x = (x_1, x_2)$  to maximize  $u(x)$ , subject to
$$\underbrace{x_1}_{g_1(x)} \geq 0 \text{ and } \underbrace{x_2}_{g_2(x)} \geq 0 \text{ and } \underbrace{m - p_1 x_1 - p_2 x_2}_{g_3(x)} \geq 0$$

## Step 1: Form the Lagrangian

- write out the thing you want to maximize
- add all expressions that must be  $\geq 0$ , multiplying the  $i$ th constraint by its own Lagrange multiplier,  $\lambda_i$ 
  - equivalently, subtract all expressions that must be  $\leq 0$ , multiplying the  $i$ th by  $\lambda_i$
  - but let's use the convention that we write all constraints in  $\geq$  form
  - eg  $p_1x_1 + p_2x_2 \leq m$  can be arranged as either requiring that  $m - p_1x_1 - p_2x_2 \geq 0$ , or that  $p_1x_1 + p_2x_2 - m \leq 0$
- so for us:

$$\begin{aligned}\mathcal{L} = & u(x_1, x_2) \\ & + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(m - p_1x_1 - p_2x_2)\end{aligned}$$

## Step 2: Take the FOC's & Slackness Conditions

- write out a *first-order condition (FOC)* for each of the original choice variables, i.e. the  $x_j$ 's: for us, this is  $\frac{\partial \mathcal{L}}{\partial x_1} = 0$  and  $\frac{\partial \mathcal{L}}{\partial x_2} = 0$
- write out a *complementary slackness condition* for each of the constraints  $g_i(x) \geq 0$ , which gives you two options (only one will work):
  - option #1, “constraint binds”:  $g_i(x) = 0$  and  $\lambda_i > 0$
  - option #2, “constraint doesn't bind”:  $g_i(x) \geq 0$  and  $\lambda_i = 0$
- idea:  $\lambda_i$  is a measure of how much the constraint limits you
- “doesn't bind” means it doesn't limit you at all: even if you set  $\lambda_i = 0$ , i.e. ignore it and optimize accordingly, you'd satisfy it anyway
- “does bind” means it *does* limit you: optimally respond by *just* satisfying it, i.e. with equality

## Step 3: Solve

- now just find the  $x_j$ 's and the  $\lambda_i$ 's that satisfy all your FOC's and slackness conditions from Step 2!
- of course, this sometimes takes some effort, but in principle it's straightforward
- **COMMENT:** this is only going to give you the right answer if preferences are **convex**

# Binding vs Non-Binding Constraints

- in our typical problem, there are 3 constraints:  $x_1 \geq 0$ ,  $x_2 \geq 0$ ,  
 $m - p_1x_1 + p_2x_2 \geq 0$
- in principle each could be binding or non-binding, leading to  $2^3 = 8$  solution possibilities
- only one will work, so you could in principle find it by trial and error
- but usually, it's better to form an educated guess on which constraints bind, and see if it works
- recall: a binding constraint is a rule you have to satisfy but would prefer not to, so you'll optimally *just* satisfy it ( $\lambda > 0$ , holds with equality)
  - eg submit midterm in  $\leq 80$  minutes; spend  $\leq \$5$  for lunch; wake up  $\leq 5$ am
  - these are likely binding: you wish you had more than 80 min, more than \$5 for lunch, could sleep later than 5am
  - so you'll optimally respond by *just* satisfying them: submit at 80 min, spend \$5 on lunch, wake up at exactly 5am



# Binding vs Non-Binding Constraints

- a non-binding constraint is a rule that doesn't actually limit you
- it's something you want to do anyway, so even if you set  $\lambda = 0$  which is like ignoring the rule, you'll satisfy it anyway
  - eg submit midterm in  $\leq 10$  hours; spend  $\leq \$1$  million on lunch; wake up by 4pm
  - presumably these are non-binding: you don't actually want to spend anywhere near 10h on the midterm, or spend anywhere near \$1 million on lunch, or sleep til 4pm
  - so even if you ignore these rules and optimize accordingly, you'll end up satisfying the rules

# Cobb-Douglas Example

- choose  $(x_1, x_2)$  to maximize  $x_1^{\frac{1}{2}} x_2^{\frac{1}{2}}$  subject to constraints  $x_1 \geq 0, x_2 \geq 0$ , and  $p_1 x_1 + p_2 x_2 \leq m$
- first, write out the Lagrangian
- then guess which constraints will bind vs not bind, and write out the corresponding equations you'd need to satisfy if your guess is right
- then, try to solve these and verify the guess

# Cobb-Douglas Example: Binding vs Non-Binding

- first, the Lagrangian:

$$\mathcal{L} = x_1^{\frac{1}{2}} x_2^{\frac{1}{2}} + \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 (m - p_1 x_1 - p_2 x_2)$$

- next, we know that the budget constraint clearly *is* binding, i.e. a real restriction
  - utility is increasing in both goods, so you'd buy  $(\infty, \infty)$  if you ignored the constraint! (which violates it)
- but choosing negative or even zero of either good gives you zero utility, clearly not optimal
  - so to maximize utility, we expect that you anyway want strictly positive quantities of both goods
  - i.e. that nonnegativity constraints are **non**-binding
- this gives us the following system of 5 equations to solve (next slide):

# Conjectured Optimality Conditions for Cobb-Douglas Example

- our educated guess is that we need to solve the following 5 equations:

|                                       |                                                                                      |
|---------------------------------------|--------------------------------------------------------------------------------------|
| FOC for $x_1$ :                       | (1) $\frac{1}{2}x_1^{-\frac{1}{2}}x_2^{\frac{1}{2}} + \lambda_1 - \lambda_3 p_1 = 0$ |
| FOC for $x_2$ :                       | (2) $\frac{1}{2}x_1^{\frac{1}{2}}x_2^{-\frac{1}{2}} + \lambda_2 - \lambda_3 p_2 = 0$ |
| non-binding constraint $x_1 \geq 0$ : | (3) $x_1 \geq 0$ and $\lambda_1 = 0$                                                 |
| non-binding constraint $x_2 \geq 0$ : | (4) $x_2 \geq 0$ and $\lambda_2 = 0$                                                 |
| binding budget constraint:            | (5) $p_1 x_1 + p_2 x_2 = m$ and $\lambda_3 > 0$                                      |

## Solution to Cobb-Douglas Example

- we want to find  $(x_1, x_2, \lambda_1, \lambda_2, \lambda_3)$  to solve (1)-(5) (in terms of  $p_1, p_2, m$ )
- from (3) and (4), we already have  $\lambda_1 = \lambda_2 = 0$
- plugging these into (1) and (2) and rearranging a bit,

$$(1a) \frac{1}{2}x_1^{-\frac{1}{2}}x_2^{\frac{1}{2}} = \lambda_3 p_1 \text{ and } (2a) \frac{1}{2}x_1^{\frac{1}{2}}x_2^{-\frac{1}{2}} = \lambda_3 p_2$$

- compare to get useful relationship: (1a)/(2a) yields
$$\frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow p_1 x_1 = p_2 x_2$$
  - note this is precisely our equal bang-per-buck condition!
- from (5), above was legit (we didn't divide by zero) since  $\lambda_3 > 0$ , and also  $p_1 x_1 + p_2 x_2 = m$
- solving,  $x_1 = \frac{m}{2p_1}$  and  $x_2 = \frac{m}{2p_2}$ ; then plug into (1a) or (2a) to get
$$\lambda_3 = \frac{1}{2\sqrt{p_1 p_2}}$$

## Lecture 4 Example 4, Revisited

- recall Example 4: maximize  $\sqrt{x_1} + x_2$ , subject to  $x_1 \geq 0$ ,  $x_2 \geq 0$ , and  $x_1 + 4x_2 \leq m$  (i.e.  $p_1 = 1$ ,  $p_2 = 4$ )
- we found that if  $m > 4$ , the consumer buys  $(x_1, x_2) = (4, \frac{m-4}{4})$
- but if  $m < 4$ , this gives a negative quantity of good 2, so go for next-best corner:  $(x_1, x_2) = (m, 0)$
- if we practice solving this problem with Lagrangians, our educated guess is:
  - if  $m > 4$ , the budget constraint binds, but non-negativity constraints do not
  - if  $m < 4$ , the budget constraint and non-negativity constraint on good 2 both bind
- let's try it with  $m = 3$ , where intuitive method gave  $(3, 0)$

## Lecture 4 Example, Revisited

- the Lagrangian is:

$$\mathcal{L} = \sqrt{x_1} + x_2 + \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 (3 - x_1 - 4x_2)$$

- the FOC's for  $x_1$  and  $x_2$  are:

$$x_1 : \frac{1}{2\sqrt{x_1}} + \lambda_1 - \lambda_3 = 0 \quad (1)$$

$$x_2 : 1 + \lambda_2 - 4\lambda_3 = 0 \quad (2)$$

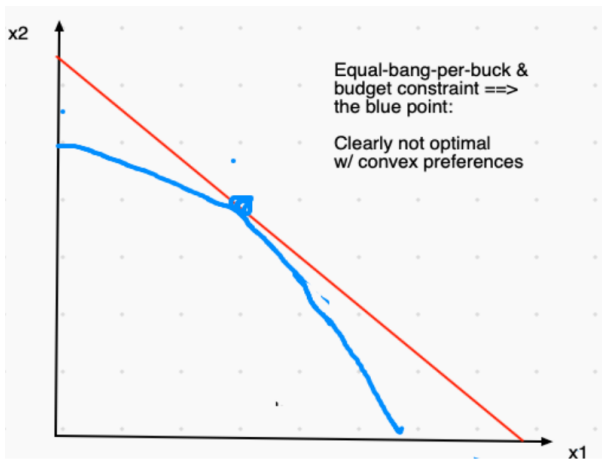
- we guessed that the  $x_1 \geq 0$  constraint doesn't bind, so try a solution with  $\lambda_1 = 0$  and  $x_1 \geq 0$  (3)
- we guessed that the  $x_2 \geq 0$  constraint does bind, so try a solution with  $\lambda_2 > 0$  and  $x_2 = 0$  (4)
- budget constraint always binds, so  $x_1 + 4x_2 = 3$  and  $\lambda_3 > 0$  (5)

# Solving

- plug (4), namely  $x_2 = 0$ , into (5), namely  $x_1 + 4x_2 = 3$ , to get  $x_1 = 3$
- we already have what we care about:  $x_1 = 3, x_2 = 0$ , which exactly spends all your money ( $x_1 + 4x_2 = 3$ ) and obeys nonnegativity constraints
- but if you wish, you can get the  $\lambda$ 's: we have  $\lambda_1 = 0$  from (3), then plug  $x_1 = 3$  and  $\lambda_1 = 0$  into (1) to get  $\lambda_3 = \frac{1}{2\sqrt{3}}$ , then plug this into (2) to get  $\lambda_2 = \frac{2}{\sqrt{3}} - 1$
- **note:** had you wrongly guessed that neither nonnegativity constraint binds, you would have replaced (4) on previous slide with  $\lambda_2 = 0$  and  $x_2 \geq 0$ ; then, with  $\lambda_2 = 0$  from (4) and  $\lambda_1 = 0$  from (3), (1) becomes  $\frac{1}{2\sqrt{x_1}} = \lambda_3$  and (2) becomes  $1 = 4\lambda_3$ ; together, this imply  $x_1 = 4$ ; plugging this into (5) on previous slide gives  $x_2 = -\frac{1}{4}$ , which violates a constraint



# Concave (Addict) Preferences



## Example 5

- assume  $u(x_1, x_2) = x_1^2 + x_2$  and  $p_1 = 2$  and  $p_2 = 1$
- what would you buy at  $m = 1$ ? what about  $m = 5$ ?
- solution: here  $MU_1 = 2x_1$  (\*increasing\* not diminishing returns!) and  $MU_2 = 1$
- so  $\frac{MU_1}{MU_2} = 2x_1$ , which **rises** as  $x_1 \uparrow$  and  $x_2 \downarrow$
- these are additive concave preferences!
- so a corner solution is best: either  $\left(\frac{m}{p_1}, 0\right)$  or  $\left(0, \frac{m}{p_2}\right)$
- at  $m = 1$ , either  $\left(\frac{1}{2}, 0\right)$  or  $(0, 1)$ : the latter is best (utility 1 rather than  $\frac{1}{4}$ )
- at  $m = 5$ , either  $\left(\frac{5}{2}, 0\right)$  or  $(0, 5)$ : the former is best (utility 6.25 instead of 5)

## Extension: More than Two Goods?

- assume convex preferences
- clearly, the Lagrangian method is already set up to handle any # goods and any # constraints, just make sure you have a FOC and a nonnegativity constraint for *each*  $x_i$
- the intuitive method also generalizes: if there are say 3 goods, an optimal interior bundle  $(x_1, x_2, x_3)$  solves:
  - equal bang-per-buck across goods:  $\frac{MU_1}{p_1} = \frac{MU_2}{p_2} = \frac{MU_3}{p_3}$
  - budget line:  $p_1x_1 + p_2x_2 + p_3x_3 = m$
  - this gives you effectively 3 equations in 3 unknowns, which you can solve
  - usual “corner solution” caveats apply

## Extension: Neither Convex Nor Concave?

- what about the 2-good case, where preferences are sometimes convex and sometimes concave?
- eg if as  $x_1 \uparrow$  and  $x_2 \downarrow$ ,  $\frac{MU_1}{MU_2}$  sometimes rises and sometimes falls, depending on the specific point?
- basically, this just gives you more candidates for the optimal point (see PS2):
  - *all* points on the budget line where goods 1 and 2 have equal bang-per-buck (there may now be several), and where preferences are convex
  - and the two corners  $\left(\frac{m}{p_1}, 0\right)$  and  $\left(0, \frac{m}{p_2}\right)$
- compare utilities for all of them

# Method of Substitution

- this is in some sense the “lowest-tech” option, and doesn’t generalize to more than 2 goods
- but there are examples where it’s the fastest way to solve a 2-good problem
- idea: suppose you want to choose  $(x_1, x_2)$  to maximize  $u(x_1, x_2)$  subject to  $p_1x_1 + p_2x_2 = m$
- step 1: solve the budget line for one good in terms of the other, eg  $x_1 = (m - p_2x_2)/p_1$
- step 2: use this to write utility as a function of one good only: eg

$$U(x_1) = u\left(x_1, \frac{m - p_1x_1}{p_2}\right) \text{ or } U(x_2) = u\left(\frac{m - p_2x_2}{p_1}, x_2\right)$$

- step 3: go back to high school calculus to maximize your function of one variable!
- step 4: then use the budget line to get the other variable
- NOTE: convex preferences  $\Leftrightarrow$  **concave**  $U(x_1)$  (and  $U(x_2)$ )

# Demand Functions

## Definitions

The demand functions  $x_1(p_1, p_2, m)$  and  $x_2(p_1, p_2, m)$  are the optimal quantities of goods 1 and 2, given prices  $p_1, p_2, m$ .

- in other words, the solution to the problem we've been solving for 2.5 classes

# Comparative Statics of Demand Functions

- we now classify goods depending on how demand changes with:
  - income
  - own price
  - other good price

# Normal vs Inferior Goods, and Luxury vs Necessity Goods

- good  $i = 1, 2$  is *normal* if demand increases as income increases:  
 $\partial x_i / \partial m > 0$ 
  - if the % change in demand  $>$  % change in income, it's called a *luxury good*
  - if the % change in demand  $<$  % change in income, it's called a *necessity good*
- good  $i = 1, 2$  is *inferior* if demand decreases as income increases:  
 $\partial x_i / \partial m < 0$
- examples?



# Income Elasticity of Demand

- *income elasticity of demand* is % change in demand divided by % change in income:

$$\eta_{x_1} = \frac{\% \Delta x_1}{\% \Delta m} = \frac{\partial x_1}{x_1} / \frac{\partial m}{m} = \frac{\partial x_1}{\partial m} \frac{m}{x_1}$$

- a good is luxury if its income elasticity exceeds 1, and a necessity good if the elasticity is smaller than 1
- example: if I gave you the demand function  $x_1(p_1, p_2, m) = m^2 / p_1$ , you would find:

$$\eta_{x_1} = \frac{\partial x_1}{\partial m} \cdot \frac{m}{x_1} = \frac{2m}{p_1} \cdot \frac{m}{m^2 / p_1} = 2$$

and conclude that it is a luxury good.

# Giffen Goods vs Ordinary Goods

## Definition

A good is an ordinary good if demand moves the opposite direction to price, and a Giffen good if demand moves the same direction as price.

- note: the Giffen condition is weird!, and Giffen goods are extremely uncommon
- Mathematical test: good  $i = 1, 2$  is ordinary if  $\partial x_i / \partial p_i < 0$ , and Giffen if  $\partial x_i / \partial p_i > 0$
- comprehension test: how do we know that a good with demand function  $x_1(p_1, p_2, m) = \frac{m}{2p_1}$  is ordinary?

# Complements vs Substitutes

- recall that *perfect substitutes* means you will trade one for the other at a constant rate
- *substitutes* is the weaker version of this: you are willing to trade one for the other, but the rate may vary
- more precisely, good 1 is a substitute for good 2 if  $\partial x_1 / \partial p_2 > 0$ : this means that if good 2 becomes more expensive, you will start *substituting* good 1 for good 2: i.e., buy more good 1 [and less good 2]. Similarly, good 2 is a substitute for good 1 if you buy more good 2 when good 1 becomes more expensive:  $\partial x_2 / \partial p_1 > 0$

# Complements vs Substitutes

- similarly, *perfect complements* are goods that you want to consume in fixed proportions; if you get more of one good, it is **only** useful if you get a proportional increase in the other
- *complements* is the weaker version of this: goods that go together, so that if one gets more expensive (presumably making you buy less), demand for the OTHER (“accompanying”) good falls as well
- formally, good 1 is a complement to good 2 if  $\partial x_1 / \partial p_2 < 0$ , and good 2 a complement to good 1 if  $\partial x_2 / \partial p_1 < 0$

# Space for Pictures

# New Topic: Income + Substitution Effects

# Income vs Substitution Effect: Idea

- next up: we are going to look at how a price increase. say in  $p_1$ , affects demand
- an increase in  $p_1$  has 2 effects
- **income effect:** you are effectively poorer than before, since higher prices reduce purchasing power (“real wealth”). How much does this affect demand?
- **substitution effect:** good 1’s bang-per-buck has now fallen, compared to good 2’s, which should nudge you to reallocate some \$ from good 1 to good 2. How much does this—the desire to substitute away to the other good—affect demand?

# Mathematically

- the goal is to decompose the total change into the two effects (Slutsky equation):

$$\text{total change} = \text{income effect} + \text{substitution effect}$$



## More Details

- suppose  $p_1$  increases to  $p'_1 > p_1$
- substitution effect: how much does  $x_1$  change due *only to reoptimization at new prices*
  - imagine (hypothetically) that you get a subsidy (the “CV”, coming up), enough to pick a point on original indifference curve: how much does good 1 demand still fall?
  - side note: it *will* fall, or at least can't rise: substitution effect is ALWAYS weakly negative
- income effect: how much does  $x_1$  change due to the fall in real wealth, i.e. when prices change but nobody actually gives you the hypothetical subsidy CV?

# Expenditure Minimization Problem

- we will solve a more general problem now, which will also give us income and substitution effects
- this is known as a **dual** problem to the utility maximization exercise we've been studying, due to a close mathematical connection

# Expenditure Minimization Problem

- consider the following problem: find the minimum expenditure (= amount of money) needed *to get a target utility level,  $u^*$* :

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2 \text{ s.t. } u(x_1, x_2) \geq u^*$$

- so instead of giving you a set income  $m$  and asking what you should buy (and how happy it makes you), this asks how much money you **need** in order to get a particular level of happiness
- the solution – the optimal bundle  $(x_1^c, x_2^c)$  as a function of  $(p_1, p_2, u^*)$  – is called the *Hicksian* (“compensated”) *demand bundle*
- the *value* of the problem, i.e.  $p_1 x_1^c(p_1, p_2, u^*) + p_2 x_2^c(p_1, p_2, u^*)$ , is called the *expenditure function*  $e(p_1, p_2, u^*)$

# Solving the Expenditure Minimization Problem

- intuitive solution: if  $(x_1^c, x_2^c)$  is interior, it solves two conditions:

$$(1) \frac{MU_1}{MU_2} = \frac{p_1}{p_2}, \text{ and } (2) u(x_1, x_2) = u^*$$

- idea: you still want to allocate your \$ optimally, but now replace the budget constraint with the utility constraint
- if you instead want to use a Lagrangian, then (excluding nonnegativity constraints) it is

$$\mathcal{L} = -p_1x_1 - p_2x_2 + \lambda (u(x_1, x_2) - u^*)$$

## Example

- assume  $u(x_1, x_2) = x_1 x_2$
- for compensated demand, condition (1) on the previous slide gives

$$\frac{x_2}{x_1} = \frac{p_1}{p_2} \Rightarrow p_1 x_1 = p_2 x_2$$

and (2) gives

$$x_1 x_2 = u^*$$

- together, these are solved by  $(x_1^c, x_2^c) = \left( \sqrt{\frac{p_2}{p_1} u^*}, \sqrt{\frac{p_1}{p_2} u^*} \right)$
- and the expenditure function is
$$e(p_1, p_2, u^*) = (p_1, p_2) \cdot (x_1^c, x_2^c) = 2\sqrt{p_1 p_2 u^*}$$
- eg at  $p_1 = p_2 = 1$ , if you want utility 5, you would need income  $2\sqrt{5}$  to make this happen, and you'd then buy the bundle  $(\sqrt{5}, \sqrt{5})$

# Application: Substitution Effect and Compensating Variation

- definition: the *compensating variation* ( $CV$ ) is the wealth subsidy needed to compensate for a price increase, so that (if you optimize) you can exactly maintain initial utility,  $u^0$
- and the *substitution effect* asks how much demand would change in response to a price increase, if you received the subsidy  $CV$
- to calculate these, supposing  $p_1 \uparrow$  to  $p'_1$ : the compensating variation is

$$CV = e(p'_1, p_2, u^0) - m$$

- and the substitution effect (change in good 1) is

$$x_1^c(p'_1, p_2, u^0) - x_1^c(p_1, p_2, u^0)$$

# Application Example

- suppose  $u(x_1, x_2) = x_1 x_2$
- Initially,  $p_1 = p_2 = 1$  and  $m = 10$ , but then  $p_1$  increases to 4
- calculate the change in demand for good 1 due to income and substitution effects
- and calculate the CV

## Application Example

- by the magic rule for Cobb Douglas preferences, this person spends half their income on each good
- so at  $p_1 = p_2 = 1$  and  $m = 10$ , this means they initially buy 5 units of each good, for utility  $u^0 = 25$
- substitution effect: from 3 slides up,  $x_1^c(p_1, p_2, u^*) = \sqrt{\frac{p_2}{p_1} u^*}$ ; so the substitution effect is

$$x_1^c(\underbrace{4, 1, 25}_{(p'_1, p_2, u^0)}) - x_1^c(\underbrace{1, 1, 25}_{(p_1, p_2, u^0)}) = \sqrt{\frac{1}{4} \cdot 25} - \sqrt{\frac{1}{1} \cdot 25} = -2.5$$

- interpretation: if this person is compensated to maintain original utility, they will reoptimize in response to the new price ratio by decreasing good 1 consumption by 2.5 units
- from 3 slides up,  $e(p_1, p_2, u^*) = 2\sqrt{p_1 p_2 u^*}$ , and so

$$CV = e(p'_1, p_2, u^0) - e(p_1, p_2, u^0) = 2\sqrt{4 \cdot 1 \cdot 25} - 2\sqrt{1 \cdot 1 \cdot 25} = \$10$$



## Application Example

- interpretation: this person needs \$10 to compensate them for the price change
- note that the initial (5,5) now costs  $(5, 5) \cdot (4, 1) = \$25$  (it initially cost  $(5, 5) \cdot (1, 1) = \$10$ )
- but this person only needs \$10, not \$15, to maintain old utility
- income effect: with no subsidy, this person optimally spends half their initial \$10 on good 1, so at new price \$4,  $4x_1^f = 5 \Rightarrow x_1^f = 1.25$ ; from previous slide,  $x_1^c = 2.5$
- so income effect is  $x_1^f - x_1^c = -1.25$
- putting it all together: when  $p_1$  increases from 1 to 4, good 1 consumption falls from 5 to 1.25: 2.5 units of this are due to reoptimization (substitution effect), and 1.25 units of this are due to income effect

## Recap: Compensated Demand & Expenditure

- the Hicksian (“compensated”) demand bundle  $(x_1^c, x_2^c)$ , a function of  $(p_1, p_2, u^*)$ , is the bundle that most efficiently=cheaply achieves target utility  $u^*$  given prices  $p_1$  and  $p_2$
- its cost  $p_1 x_1^c + p_2 x_2^c$  is called the expenditure function: how much money do you need to get utility  $u^*$
- to find the compensated bundle,
  - (1) allocate \$ optimally between goods (same as before, usually this is to equate bang-per buck);
  - (2) choose a bundle that gives utility  $u^*$  (instead of picking a point on the budget line)

## Recap: Substitution Effect and Compensating Variation

- definition: the *compensating variation* ( $CV$ ) is the wealth subsidy needed to compensate for a price increase, so that (if you optimize) you can exactly maintain initial utility,  $u^0$
- and the *substitution effect* asks how much demand would change in response to a price increase, if you received the subsidy  $CV$
- to calculate these, supposing  $p_1 \uparrow$  to  $p'_1$ : the compensating variation is

$$CV = e(p'_1, p_2, u^0) - m$$

- and the substitution effect (change in good 1) is

$$x_1^c(p'_1, p_2, u^0) - x_1^o$$

# Shepard's Lemma

*The expenditure function and Hicksian demand functions satisfy the following relationship:*

$$\frac{\partial e(p, u)}{\partial p_i} = x_i^c(p, u)$$

- earlier we learned that if you already know  $x_1^c(p, u^*)$  and  $x_2^c(p, u^*)$ , calculate their cost to get the expenditure function  $e(p, u)$
- this says that conversely if you know the expenditure function, you can easily get the  $x_i^c$ 's

# Proof of Shephard's Lemma

- by definition, expenditure is the cost of the Hicksian (compensated) bundle:

$$e(p_1, p_2, u) = p_1 x_1^c(p_1, p_2, u) + p_2 x_2^c(p_1, p_2, u)$$

- differentiate this w.r.t.  $p_1$  :

$$\frac{\partial e(p_1, p_2, u)}{\partial p_1} = x_1^c(p_1, p_2, u) + \underbrace{p_1 \frac{\partial x_1^c}{\partial p_1} + p_2 \frac{\partial x_2^c}{\partial p_1}}_{\text{second-order effects}}$$

- by the “envelope theorem”, only the direct effect matters
- optimality FOC's imply that the second-order effects vanish
- QED!

# Envelope Theorem: Illustration in Single-Variable Case

- suppose you choose  $x$  to minimize (or maximize) a function  $f(x, p)$ , where  $p$  is a parameter
- and you get an optimal solution  $x^*(p)$
- then  $\frac{d}{dp} f(x^*(p), p) = f_x \cdot \frac{d}{dp} x^*(p) + f_p$
- but  $f_x = 0$  via optimization process!
- so this is simply  $f_p$
- envelope theorem says this holds with multiple variables as well